Behavioral Speciation in the Neuroevolution of Recurrent Neural Net Foragers driven by Environmental Homogeneity

Damian R. Sowinski[1,2]
[1]Department of Physics & Astronomy, University of Rochester, NY USA
[2]S.E.T.I. Institute, Menlow Park, CA USA drsowinski@gmail.com

Abstract

Neuroevolution of foraging recurrent neural networks is investigated using the Layered Environment to Agent Foraging Simulator (LEAFS), an optimized agent-based framework. Speciation to high fitness behavioral phenotypes is observed across genetically diverging populations; three distinct locomotive modalities are cataloged. During initial stages of evolution, the discovery of higher fitness behaviors, depends weakly on environmental homogeneity. In more-homogeneous environments, discovery is punctuated yet the effect on the population is gradual. In lower-homogeneity environments both individual fitness and population fitness increase over multiple rapid steps.

Submission type: **Full Paper**

Neuroevolution is a computational approach towards understanding how natural selection optimizes neural architecture Miikkulainen (2025). Coupling network evolution to fitness evaluation in simulated environments, it provides a controlled laboratory for studying the emergence of adaptive behavior. This work studies populations of recursive neural network (RNN) agents which need to forage to pay their metabolic costs. Neuroevolution allows these populations to explore the fitness landscape, find informative representations of their environments, and optimize energy harvesting. Both convergent and divergent evolution are observed through the speciation of agents into distinct behavioral phenotypes across environments of differing resource homogeneities.

Elman networks, introduced in 1990, are arguably the simplest RNNs Elman (1990). Their hidden layer feeds back across timesteps, endowing agents with memory of recent sensory history. Evolution in populations of foraging agents controlled by these RNNs began soon after, on small, discrete grids Nolfi et al. (1994). Parallel developments evolved continuous-time recurrent neural networks (CTRNNs) for chemotaxis and sequential foraging Beer and Gallagher (1992); Beer (1995), and concurrently evolved sensor morphology alongside recurrent controllers for visually-guided behavior Harvey et al. (1994). In each case, internal representations of environmental structure emerged without being prescribed. These approaches were extended to richer settings: homing navigation to an energy source was evolved on physical robots, with emergent topographic neural maps arising without design Floreano and Mondada (1996); evolving neural learning rules in a bumblebee foraging model recovered documented behavioral phenotypes including risk aversion and probability matching Niv et al. (2002). Neuroevolution of foraging reliably produces biologically interpretable strategies.

Since Charnov's marginal value theorem Charnov (1976), optimal foraging theory has framed energy harvesting as an optimization problem shaped by environmental pressure. The physics of random search has been equally productive: in sparse, unpredictable environments, superdiffusive walks outperform Brownian motion Viswanathan et al. (1999, 2011); James et al. (2008), while intermittent strategies — alternating ballistic motion with local search — optimize encounter rates with hidden targets Bénichou et al. (2005). These analytically derived strategies provide a baseline against which evolved ones can be compared. Physical limits on sensory precision further constrain achievable strategies Berg and Purcell (1977), particularly how much information about an agent's local environment is needed for a strategy to be successful Mattingly et al. (2021); Sowinski et al. (2023); Rode et al. (2024).

A recurring observation is that independent evolutionary runs converge on functionally similar behaviors via genetically distinct solutions — a computational echo of convergent evolution. Mouret and Doncieux showed rigorously that genotypic and behavioral diversity are decoupled in evolutionary robotics Mouret and Doncieux (2012). The fitness landscape of neuroevolution problems is generically multimodal Rodrigues et al. (2020); Miikkulainen (2025), and Lehman and Stanley showed that explicitly rewarding behavioral novelty can outperform objective-based search in such landscapes Lehman and Stanley (2011).

This work examines the neuroevolution of foraging agents

©2026 Damian R Sowinski. Published under a Creative Commons Attribution 4.0 International (CC BY 4.0) license.

with contiguous sensor arrays and Elman RNN controllers across environments of varying resource homogeneity, Ξ, described in section II. Section III describes the simulations, carried out using the Layered Environment to Agent Foraging Simulator (LEAFS) Sowinski et al. (2023). The fitness landscape is found to have many maxima, with populations across independent histories discovering different ones. Section IVshows that behavioral phenotypes fall into three categories despite divergent inter-history genomes. The transition dynamics to high-fitness strategies depend qualitatively on Ξ: homogeneous environments yield single-burst transitions at the individual level but slow population fitness-gradient climbs, while heterogeneous environments produce multi-stage, punctuated progressions reminiscent of punctuated equilibrium in biological evolution Eldredge et al. (1972). Prospects for future work are discussed in section V.

Theory

Consider the model of Sowinski et al. (2023): an environment populated by foraging agents and peppered with resources. The 2-dimensional world has an energy influx Γ generating resources of energy ϵ that decay at a rate τ_E^{-1}. This is a homogeneous point process resulting in a mean resource energy density $\rho^{eq} = \Gamma\tau_E$. The relevant length scale is accordingly the mean resource spacing, $\ell_E^2 = \epsilon/\rho^{eq}$.

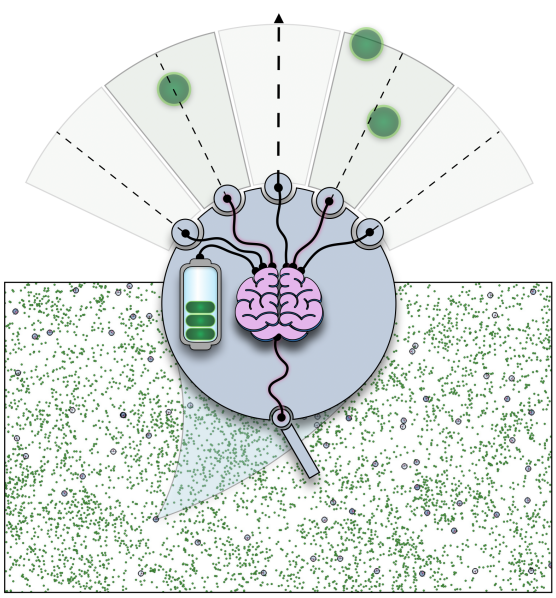

Figure 1: (Background) A system state in LEAFS. Resources appear in green and foragers in blue; the forager population exhibits clear spatial correlations with the resource field. (Foreground) Basic architecture of a foraging agent. Sensors detect nearby resources and send signals to an RNN that controls the agent's movement. Harvested resources replenish the agent's fuel supply, which is consumed by metabolic activity.

Environmental homogeneity is defined relative to the agent size. For a given resource spacing, agents larger than that spacing encounter a homogeneous energy field, while agents smaller than it encounter a heterogeneous one. Accordingly, homogeneity is quantified by the dimensionless ratio

$$\Xi = \frac{\ell_A}{\ell_E}. \tag{1}$$

The language herein refers to environments with $\Xi \geq 0.8$ as homogeneous, and those with smaller Ξ as heterogeneous.

The agent degrees of freedom are its position $\boldsymbol{r}$, orientation θ, fuel supply s, and neural state $\boldsymbol{h}$. Agents move through their environment at constant speed v,

$$\frac{d\boldsymbol{r}}{dt} = v\hat{\boldsymbol{n}}, \tag{2}$$

where $\hat{\boldsymbol{n}} = \hat{\boldsymbol{n}}(\theta)$ is the unit vector along the agent's orientation. The agent's fuel supply is consumed by basal metabolism μ_0 and replenished by harvesting resources,

$$\frac{ds}{dt} = -\mu_0 + h. \tag{3}$$

Here h is the harvest rate: the fuel supply increases by ϵ, up to a maximum S, whenever a resource falls within the agent's collection radius r. When the fuel supply surpasses a reproduction threshold, $s(t) \geq s_{\text{rep}}$, the agent splits into two daughters with equally divided fuel. The daughters are placed antipodally at a distance r from the parent's centroid, at a uniformly random angle.

The key difference from Sowinski et al. (2023) is how agents process sensory information. Rather than using a biologically motivated decision tree, agents are equipped with a single hidden layer recurrent neural network. Sensors on the agent's body each sweep a cone of limited range: N_S sensors are fixed at angular positions θ_I relative to the agent's orientation, $I = 1, 2, \ldots, N_S$. Each sensor has a field of view of angular width ϕ out to sensing radius R. An empty field of view gives an activation of 0; one resource gives 1; two or more give 2. Sensors can be interpreted physically as photo- or chemoreceptors with a rudimentary ability to distinguish local environmental states.

The agent's neural state consists of N_h fully connected perceptrons and evolves as

$$\boldsymbol{h}(t + \Delta t) = \tanh(\boldsymbol{W}_{hh} \cdot \boldsymbol{h}(t) + \boldsymbol{W}_{hx} \cdot \boldsymbol{x}(t) + \boldsymbol{b}_h)\,, \tag{4}$$

with the hyperbolic tangent applied pointwise. The neural state determines the probability distribution over effector states,

$$\{p(\alpha)\} = \text{softmax}(\boldsymbol{W}_{\alpha h} \cdot \boldsymbol{h}(t + \Delta t) + \boldsymbol{b}_\alpha)\,. \tag{5}$$

Here the $\boldsymbol{W}$'s are weight matrices and the $\boldsymbol{b}$'s are biases. During reproduction, daughters' weights and biases are perturbed from the parent's: $(\boldsymbol{W}, \boldsymbol{b})_{\text{daughter}} = (\boldsymbol{W}, \boldsymbol{b})_{\text{parent}} + (\delta\boldsymbol{W}, \delta\boldsymbol{b})$, where entries of $\delta\boldsymbol{W}$ and $\delta\boldsymbol{b}$ are drawn independently from a Gaussian with mean 0 and standard deviation δ. These mutations drive evolutionary exploration of genomic space. The input vector $\boldsymbol{x}(t)$ consists of the sensor activations, the agent's fuel supply, and a binary flag indicating whether a resource was consumed in the last timestep.

The action repertoire is $\alpha \in \{\omega, 0, -\omega\}$, a set of rotational speeds that steer the agent via

$$\frac{d\theta}{dt} = \alpha + \sigma\eta, \tag{6}$$

where $\sigma \geq 0$ is the tumble parameter. The regimes $\sigma > 1$ and $\sigma \ll 1$ correspond to tumbling and ballistic motion, respectively; this work is confined to the ballistic regime.

Experiments

	Environment	
Γ	Energy influx	10^{-3}
τ_E	Environmental timescale	10
ρ^{eq}	Resource energy density	10^{-2}
A	Simulation area	$10^2 \times 10^2$
Ξ	Homogeneity	0.1–2
	Agent	
τ_A	Agent timescale	0.08
ℓ_A	Agent length scale	2
R	Sensing radius	3.24
N_h	Hidden layer size	3
s_{rep}	Reproduction threshold	0.8
σ	Tumble parameter	10^{-1}
δ	Mutation rate	0.2
N_S	Number of sensors	5
S	Maximum fuel supply	1
μ_0	Metabolism	1
ω	Rotation speed	10π
	Numerics	
Δt	Timestep	0.02

Initial experiments on random sensor placement did not exhibit appreciable fitness changes on the simulation history timescales herein. Accordingly, this work focuses on the neuroevolution of agents with contiguous sensor arrays. Five sensors, each with field of view $\pi/5$, are placed at angular locations $\{0, \pm\pi/5, \pm 2\pi/5\}$. Contiguity is crucial: moving past a resource causes nearby sensors to fire sequentially, creating temporal correlations that reflect the geometry of the environment. For the central sensor to cover the full sweep of the agent's collection area, the sensing radius is set to $R/r = 1/\sin(\pi/10) \approx 3.24$.

Agents are given a very simple architecture. Data from the five sensors, as well as the current health and a binary tag describing whether a resource was harvested in the last timestep, feed forward to three hidden units. These three are recursively connected to themselves, and feed forward to the output layer. The output consists of the three units responsible for the softmax action distribution. This gives a total of $7 \times 3 + 3! + 3 \times 3 = 36$ weights, and 6 biases. These 42 parameters form the *genome* of the agents.

Simulations are run on the BlueHive HPC cluster at the University of Rochester using an in-house MATLAB codebase: LEAFS. The environment is $100 \times 100\,\ell_A^2$ with periodic boundary conditions. LEAFS optimizes nearest-neighbor searches by discretizing the environment into $5 \times 5\,\ell_A^2$ cells, reducing resource-agent pair comparisons by 97.5%. A timestep of $\Delta t = 0.02\,\tau_A$ is used; because the RNN updates discretely while agent translation and rotation are continuous, a forward Euler scheme integrates the degrees of freedom. Simulations span homogeneity values from 0.1 to 2; to hold the environmental timescale fixed, Ξ is varied by adjusting the resource energy ϵ.

For each value of Ξ, ten seed agents are instantiated with $s = S/2$, their genomes drawn from the Gaussian matrix ensemble. Each agent is placed in an independently instantiated environment, and the resulting time series is referred to as a *history*. Agent lifetimes within a history are recorded and a *longest-lived hall-of-fame* (HOF) is maintained: as agents live and die, the five longest-lived genomes are kept.

Each history is divided into 100 *epochs*. An epoch lasts either a fixed number of timesteps ($5 \times 10^4\,\Delta t$) or until an extinction, whichever comes first. In the former case, all foragers are removed, the empty environment is repopulated with genomes drawn uniformly from the HOF, and the HOF is reset. The long epoch length ensures the population explores large deviations from the previous HOF. If an extinction occurs instead, the HOF genomes are perturbed as in a reproduction event before repopulating; this helps populations escape non-viable minima in the fitness landscape.

Consequently, many early epochs are short, lasting little more than a single agent lifetime. In most histories, populations eventually discover a behavioral strategy that significantly increases individual lifetimes. There is a qualitative inverse trend between the fraction of populations that discover such a strategy and the environmental homogeneity, consistent with prior analysis showing that sensors become advantageous only below a threshold homogeneity et al (2026).

After each epoch, the HOF is recorded along with multiple agent and environment statistics. To estimate the joint distributions, the HOF is sampled at fixed time intervals. For each sample, an environment is instantiated with the HOF population and run for a fixed number of timesteps; if an extinction occurs, the environment is repopulated with the same genomes. These runs exclude genetic drift, isolating the effect of the HOF genomes on agent-environment statistics. Sensor and effector states are recorded at each timestep

and used to populate an empirical distribution.

Results and Discussion

Evolution of behavioral strategies depends on environmental homogeneity. There is a slight inverse trend between the fraction of histories converging to a high-fitness strategy and resource homogeneity, though this trend disappears for $\Xi < 0.1$ due to limited data. At high heterogeneity, the probability of discovering a near-optimal phenotype by chance — one capable of locating widely spaced resources — becomes negligible, and populations fail to converge.

Comparing the top panels of figures 2–3 shows that in homogeneous environments the transition from short-lived to long-lived agents is rapid. In heterogeneous environments the transition proceeds through multiple stages. This increase in the number of steps is observed qualitatively across many histories. Provided the environment is not too sparse, populations can discover good-enough phenotypes occupying large volumes of genome space. These are then further refined toward more efficient behaviors over subsequent epochs.

The bottom panels of figures 2-3 show overall forager and resource counts, both normalized to the carrying capacity of the environment. For resources, this is the no-agent equilibrium

$$N_R^{\max} = \frac{\Gamma \tau_E}{\epsilon} A, \tag{7}$$

and for agents

$$N_A^{\max} = \frac{\Gamma}{\mu_0} A. \tag{8}$$

The symmetry between the two curves follows from energy balance at equilibrium,

$$\Gamma A - \epsilon \tau_E N_R^{eq} - \mu_0 N_A^{eq} = 0, \tag{9}$$

which gives

$$\frac{N_R^{eq}}{N_R^{\max}} + \frac{N_A^{eq}}{N_A^{\max}} = 1. \tag{10}$$

The bottom panels show that it in homogeneous environments reaching peak lifetime does not drive rapid population growth. Post lifetime maximization, the agent population gradually increases towards its asymptotic value. In heterogeneous environments, the discovery of effective behavioral strategies produces step-like jumps in the agent population curve. Across different histories for the same Ξ, populations whose lifetimes plateau do so at very similar values. The pink curve in figure 4 shows this, which will be discussed later. In what follows, only these converged histories are analyzed.

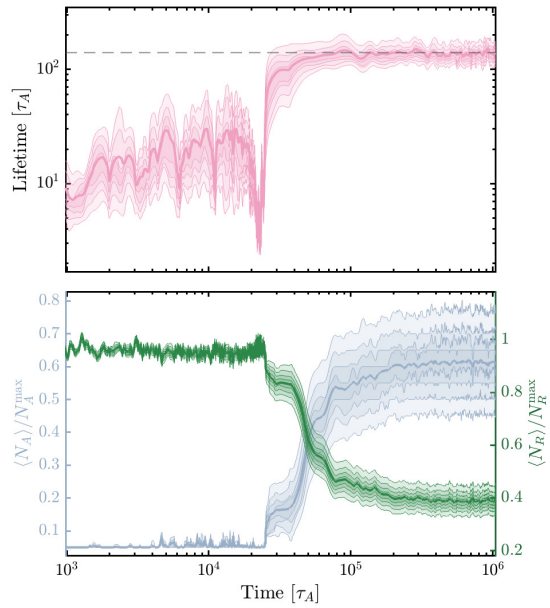

Figure 2: Evolution in a homogeneous environment, $\Xi = 0.8$. (Top) Behavioral strategies evolve, increasing average agent lifetimes in a single burst. (Bottom) Normalized average resource and agent counts. The behavioral shift produces a gradual change in population statistics. Data is collected across epochs, each time point representing a distribution over $100\tau_A$. The solid line is the median of that distribution, and filled regions are 10% quantiles.

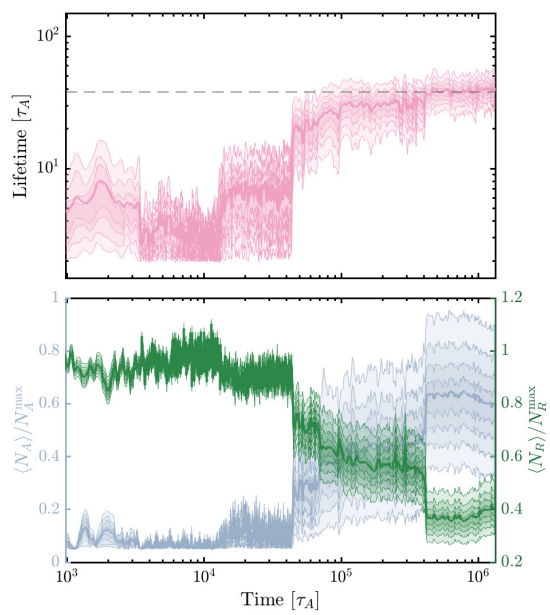

Figure 3: Evolution in a heterogeneous environment, $\Xi = 0.1$. (Top) Behavioral strategies evolve in bursts, driving multi-stage lifetime increases that are both steplike and gradual. (Bottom) Normalized average resource and agent counts. Behavioral shifts have clear analogues in the major changes in population.

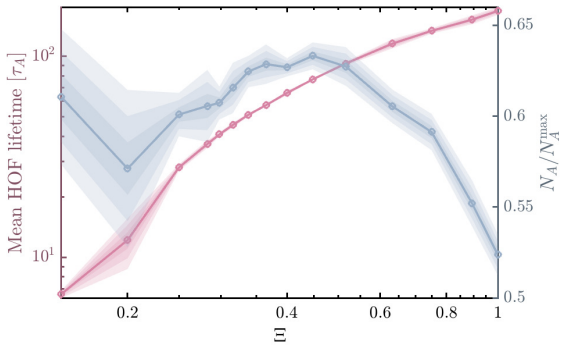

Figure 4: Converged HOF statistics as a function of environmental homogeneity Ξ. Dark lines are means across histories; shaded regions are standard deviations. Agent lifetime (pink) increases with Ξ. Normalized agent population is shown in grey.

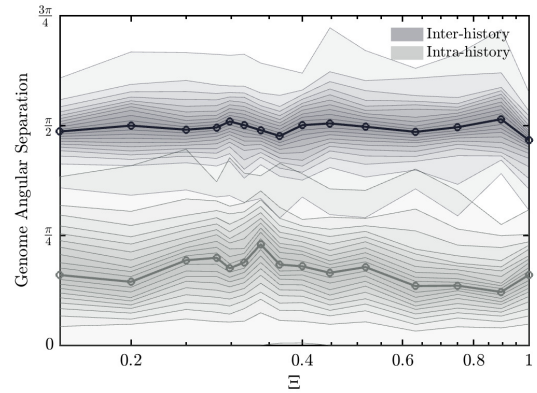

Figure 5: Angular separation of genomes within a single history (intra) and across histories (inter). Shaded regions are 5% quantile bands; dark lines are medians. Histories discover genotypically distinct solutions to survival.

Although equation 10 constrains the equilibrium number densities of resources and agents, it does not uniquely determine them. The equilibrium forager count, normalized to $N_A^{\max}$, is shown as a function of homogeneity in figure 4. Surprisingly, this normalized population fraction grows with decreasing resource homogeneity, peaks near $\Xi \approx 0.4$, and then appears to decrease. Data are sparser at low homogeneity, so the count may instead plateau. This feature is mirrored in the behavioral distributions of figure 7, discussed below.

What behaviors lead to lifetime and population growth? Qualitative analysis shows that foragers in high-lifetime histories turn toward nearby resources. Evolutionarily, RNNs are expected to learn to modulate behavior based on local environmental state to increase energy harvesting. Since the sensor array is contiguous, it is plausible that RNNs evolve contiguous internal representations of the environment — representations suited to effectors controlling a continuous rotational degree of freedom. Whether non-contiguous sensor arrays with non-overlapping fields of view can support equivalent representations is currently under investigation using longer simulation timescales.

The generic turning-toward-resources behavior is shared across all high-lifetime runs, yet this functional convergence is hidden at the level of genotypes. Genome variation within a single history is small compared to variation between histories, as expected from the epoch-based repopulation scheme. To quantify this, collapse each forager's weights and biases to a genome vector, $\{\boldsymbol{W}, \boldsymbol{b}\} \mapsto \boldsymbol{w}_i$. The angular separation between genomes i and j is

$$\Delta(\boldsymbol{w}_i, \boldsymbol{w}_j) = \cos^{-1} \frac{\boldsymbol{w}_i \cdot \boldsymbol{w}_j}{||\boldsymbol{w}_i||\,||\boldsymbol{w}_j||}, \tag{11}$$

where small angles indicate similar genomes and angles near $\pi/2$ indicate orthogonal ones. Figure 5 shows the distribution of pairwise separations at fixed Ξ. Inter-history separations lie close to $\pi/2$: genomes from different histories are nearly orthogonal. Intra-history separations are typically below $\pi/4$.

The behavioral diversity of inter-history genomes is much lower than their genomic diversity. This convergence is most apparent in the action distributions conditioned on coarse-grained sensor states. To aid visualization, define the resource gradient g as follows: binarize the sensor activations to $\bar{\boldsymbol{x}}$, then form the dot product with $\boldsymbol{w} = (-2, -1, 0, 1, 2)$, so that $g = \bar{\boldsymbol{x}} \cdot \boldsymbol{w}$ is negative when resources are predominantly to the left and positive when they are to the right. The resulting conditional distributions $P(\alpha|g)$ fall into three distinct categories shown in figure 6. Despite this behavioral distinction, figure 4 shows that all three phenotypes produce indistinguishable population-level statistics.

Beeline foragers move in straight lines, turning only when a resource appears at the periphery of their field of view. Saccadic foragers turn more erratically. Serpentine foragers zigzag continuously, never maintaining a steady heading. Classes were initially identified by inspecting the conditional distributions and hand-labeling a small number of genotypes; the remainder were labeled by a k-means clustering algorithm.

The phenotype distribution as a function of Ξ is shown in figure 7. Serpentine locomotion is rare across all environments. Serpentine foragers never move straight, always choosing to turn left or right. On average they progress toward resources, but at an effectively lower speed, implying a reduced harvest rate. One might expect this to produce shorter lifetimes, yet figure 4 shows that serpentine foragers are equally long-lived. Current work is looking at disentangling the more nuanced differences in population statistics attributable to these phenotypes.

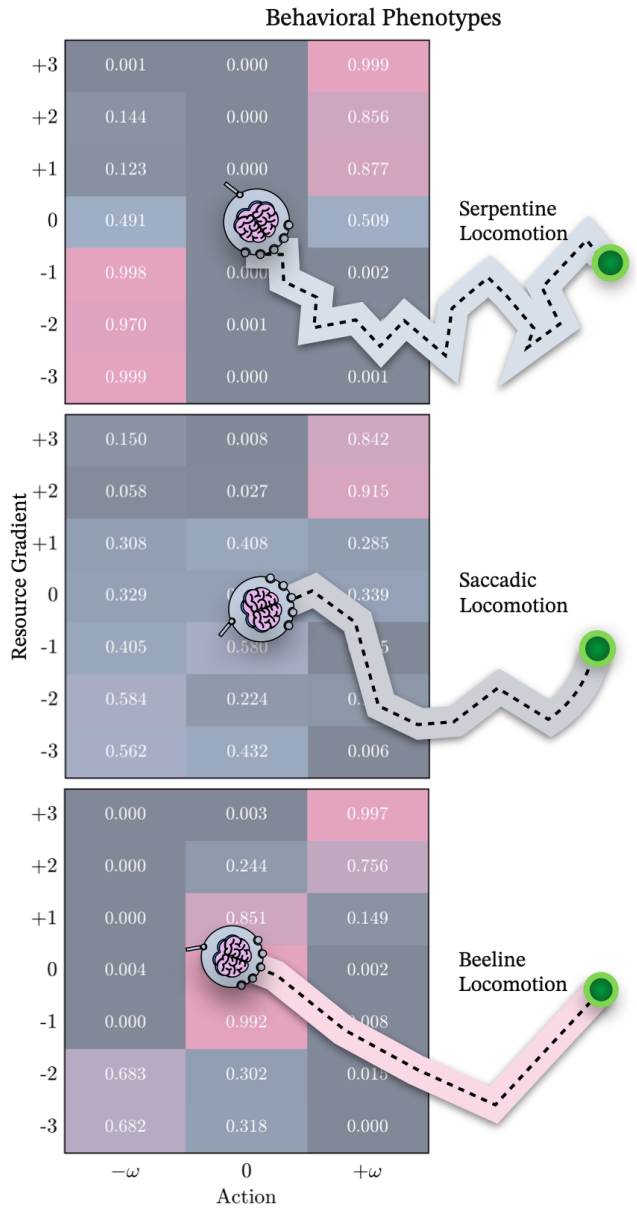

Figure 6: Three behavioral phenotypes observed across environments and histories. Background: action distributions conditioned on the nearby resource gradient, $P(\alpha|g)$. Pink corresponds to higher probability and grey to lower. Foreground: representative trajectories for each phenotype. How these relate to

Moving from serpentine to saccadic phenotypes, probability mass shifts from the extremal turning actions ($\pm\omega$) toward the null action (0). This trend continues toward beeline behavior. In sparse environments, foragers must identify resources and move toward them efficiently; time spent turning instead of translating depletes energy that could otherwise be directed toward the next harvest. Beeline ballistic

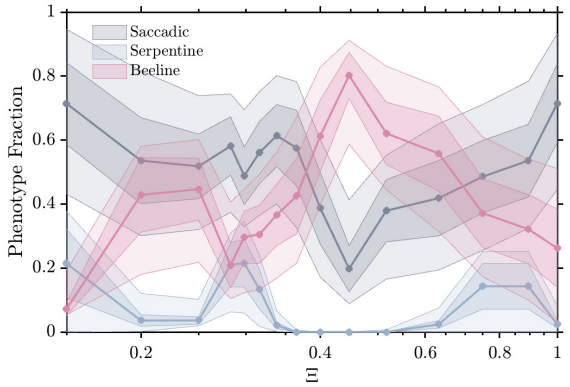

Figure 7: Phenotype fractions across the range of environmental homogeneities studied. Statistics are pooled over all histories. The dark central line is the mean; shaded regions are quartiles.

motion appears optimal by this reasoning, yet it dominates only within a narrow band of homogeneities close to the peak in figure 4. Outside this band, figure 7 suggests that saccadic locomotion is the dominant phenotype.

Conclusion

Neuroevolution's solution to the problem of survival takes on more distinguishable features as one moves down the population–phenotype–genotype hierarchy. At the population level, either equilibrium is reached or extinction occurs. At the phenotype level, a small number of locomotive strategies emerge. At the genotype level, the space is vast: many distinct neural circuits produce the same conditional distributions, and survival is a problem with many solutions.

Serpentine, saccadic, and beeline foragers, given fixed genomes, are analogous to distinct species. Pinpointing any differences in population statistics for each species is ongoing. This is the first time such behavioral speciation has been observed in the LEAFS framework, albeit across separate histories of identical environments. Future work will investigate whether speciation can also emerge within a single history, and how the information agents extract from their environments is organized into internal representations across evolutionary epochs.

Acknowledgments

Ideas do not appear fully formed from the void, but as seeds in need of water. Thanks to Jose Perez Mercador, Hiroki Sayama, Giorgi Georgev, Alice Quillen, Gourab Ghoshal, and Aidan Meador-Woodruff for their thought-provoking conversations. Many thanks to Chaitanya Murthy for expressing enthusiastic interest in initial results, without which this work would have languished in the ever-chaotic seas of a novelty-seeking mind. This work was done at the Uni-

versity of Rochester using the HPC resources at CIRC. Anthropic's Claude (Sonnet 4.6) was used to set up simulations on CIRC's BlueHive cluster, optimize older versions of LEAFS, and write analysis and plotting scripts. The Stanford Agentic Reviewer was used to review initial drafts of the paper.

References

Beer, R. D. (1995). On the dynamics of small continuous-time recurrent neural networks. *Adaptive Behavior*, 3(4):469–509.

Beer, R. D. and Gallagher, J. C. (1992). Evolving dynamical neural networks for adaptive behavior. *Adaptive behavior*, 1(1):91–122.

Bénichou, O., Coppey, M., Moreau, M., Suet, P., and Voituriez, R. (2005). Optimal search strategies for hidden targets. *Physical review letters*, 94(19):198101.

Berg, H. C. and Purcell, E. M. (1977). Physics of chemoreception. *Biophysical journal*, 20(2):193–219.

Charnov, E. L. (1976). Optimal foraging, the marginal value theorem. *Theoretical population biology*, 9(2):129–136.

Eldredge, N., Gould, S. J., et al. (1972). Punctuated equilibria: an alternative to phyletic gradualism. *Models in paleobiology*, 82:115.

Elman, J. L. (1990). Finding structure in time. *Cognitive science*, 14(2):179–211.

et al, D. S. (2026). To see or not to see, that is the question when one is hungry. unpublished, in preparation.

Floreano, D. and Mondada, F. (1996). Evolution of homing navigation in a real mobile robot. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, 26(3):396–407.

Harvey, I., Husbands, P., Cliff, D., et al. (1994). Seeing the light: Artificial evolution, real vision. *From animals to animats*, 3:392–401.

James, A., Plank, M., and Brown, R. (2008). Optimizing the encounter rate in biological interactions: ballistic versus lévy versus brownian strategies. *Physical Review E—Statistical, Nonlinear, and Soft Matter Physics*, 78(5):051128.

Lehman, J. and Stanley, K. O. (2011). Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary computation*, 19(2):189–223.

Mattingly, H. H., Kamino, K., Machta, B. B., and Emonet, T. (2021). Escherichia coli chemotaxis is information limited. *Nature physics*, 17(12):1426–1431.

Miikkulainen, R. (2025). Neuroevolution insights into biological neural computation. *Science*, 387(6735):eadp7478.

Mouret, J.-B. and Doncieux, S. (2012). Encouraging behavioral diversity in evolutionary robotics: An empirical study. *Evolutionary computation*, 20(1):91–133.

Niv, Y., Joel, D., Meilijson, I., and Ruppin, E. (2002). Evolution of reinforcement learning in uncertain environments: A simple explanation for complex foraging behaviors. *Adaptive Behavior*, 10:1.

Nolfi, S., Parisi, D., and Elman, J. L. (1994). Learning and evolution in neural networks. *Adaptive Behavior*, 3(1):5–28.

Rode, J., Novak, M., and Friedrich, B. M. (2024). Information theory of chemotactic agents using both spatial and temporal gradient sensing. *PRX Life*, 2(2):023012.

Rodrigues, N. M., Silva, S., and Vanneschi, L. (2020). A study of fitness landscapes for neuroevolution. In *2020 IEEE Congress on Evolutionary Computation (CEC)*, pages 1–8. IEEE.

Sowinski, D. R., Carroll-Nellenback, J., Markwick, R. N., Piñero, J., Gleiser, M., Kolchinsky, A., Ghoshal, G., and Frank, A. (2023). Semantic information in a model of resource gathering agents. *PRX Life*, 1(2):023003.

Viswanathan, G. M., Buldyrev, S. V., Havlin, S., da Luz, M. G., Raposo, E. P., and Stanley, H. E. (1999). Optimizing the success of random searches. *nature*, 401(6756):911–914.

Viswanathan, G. M., Da Luz, M. G., Raposo, E. P., and Stanley, H. E. (2011). *The physics of foraging: an introduction to random searches and biological encounters*. Cambridge University Press.